

SQL Query Results

Caleb Gevertz (JHED: cgevert1) - Modern Software Concepts in Python, Module 3

Generated 2026-09-20 from the PostgreSQL 'applicants' table (40,000 entries). Counts are whole numbers, percentages show two decimals, and averages show two decimals.

Note on LLM columns: llm_generated_program / llm_generated_university are filled for 10,442 of the 40,000 entries; the local LLM (about 0.75 s per entry) was run on the entries that matter for Question 9 first. Every Fall 2026 accepted PhD entry (the only entries Question 9 can select) has LLM values, so Question 9 is not affected by the entries that were not standardized.

Why Questions 8 and 9 agree: both searches found 30 entries, and the two sets of matching entries are identical. Grad Cafe's current results already give the program and the university as separate, cleanly typed fields (Module 2 rebuilt the combined 'Program, University' text only so the local LLM had something to split), so with input that tidy the model mostly copies the names back. That does not mean LLM standardization is useless: it matters for free-text or misspelled entries such as 'John Hopkins' or unusual abbreviations, where it can raise the count, and it can also lower it by mistake (I saw spelling errors such as 'Fayeletteville' and 'Ellectrical'). A difference of zero here only shows that, for these four universities, the original fields were already good enough.

Question 1: Fall 2026 applicants

Question

How many entries in the database are from applicants who applied for Fall 2026?

Result

Fall 2026 applicant count: 33,207

SQL query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026';
```

What the query does and why it answers the question

Counts the rows whose term is Fall 2026. The term is trimmed and lower-cased first so 'Fall 2026', 'fall 2026' and ' Fall 2026 ' all match.

Question 2: Percent international

Question

Among entries that provide a nationality classification, what percentage are international students?

Result

Percent international: 47.14%

SQL query

```
SELECT 100.0 * COUNT(*) FILTER (WHERE LOWER(us_or_international) = 'international')
/ NULLIF(COUNT(*) FILTER (WHERE NULLIF(TRIM(us_or_international), '') IS NOT NULL), 0)
FROM applicants;
```

What the query does and why it answers the question

The numerator counts International entries. The denominator counts every entry with a non-blank classification (American, International or Other), so blank values are left out and American and Other count as not international. NULLIF guards against dividing by zero.

Question 3: Average GPA and GRE scores

Question

What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Result

Average GPA: 3.76

Average GRE Quantitative: 165.68

Average GRE Verbal: 160.35

Average GRE Analytical Writing: 4.32

SQL query

```
SELECT AVG(gpa), AVG(gre), AVG(gre_v), AVG(gre_aw)
FROM applicants;
```

What the query does and why it answers the question

AVG ignores NULLS, so each of the four averages is computed only over applicants who supplied that particular metric. An applicant with a GPA but no GRE still counts toward the GPA average.

Question 4: Average GPA of American Fall 2026 applicants

Question

What is the average GPA of American applicants who applied for Fall 2026?

Result

Average GPA of American Fall 2026 applicants: 3.79

SQL query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
      AND LOWER(TRIM(us_or_international)) = 'american'
      AND gpa IS NOT NULL;
```

What the query does and why it answers the question

Filters to Fall 2026 entries classified as American that actually report a GPA, then averages the GPA.

Question 5: Fall 2025 acceptance percentage

Question

What percentage of Fall 2025 entries are acceptances?

Result

Fall 2025 acceptance percentage: 39.03%

SQL query

```
SELECT 100.0 * COUNT(*) FILTER (WHERE status ILIKE 'accept%') / NULLIF(COUNT(*), 0)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2025';
```

What the query does and why it answers the question

Among all Fall 2025 entries, COUNT(*) FILTER counts those whose status starts with 'accept' (case-insensitive) and divides by the total number of Fall 2025 entries.

Question 6: Average GPA of accepted Fall 2026 applicants

Question

What is the average GPA of accepted applicants who applied for Fall 2026?

Result

Average GPA of accepted Fall 2026 applicants: 3.76

SQL query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
      AND status ILIKE 'accept%'
      AND gpa IS NOT NULL;
```

What the query does and why it answers the question

Filters to Fall 2026 entries with an acceptance status that report a GPA, then averages the GPA.

Question 7: Johns Hopkins Computer Science master's entries

Question

How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Result

Johns Hopkins University Computer Science master's entries: 20

SQL query

```
SELECT COUNT(*)
FROM applicants
WHERE (program ILIKE '%john% hopkins%' OR program ~* '\yjhu\y')
      AND program ILIKE '%computer science%'
      AND degree ILIKE 'master%';
```

What the query does and why it answers the question

Uses the original downloaded program and degree fields. The program text must mention Johns Hopkins (including the 'John Hopkins' misspelling) or the standalone word JHU, and contain 'computer science', and the degree must start with 'master'.

Question 8: Fall 2026 CS PhD acceptances at four universities (original fields)

Question

How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at Georgetown, MIT, Stanford or Carnegie Mellon?

Result

Fall 2026 accepted PhD Computer Science entries (original fields): 30

SQL query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
      AND status ILIKE 'accept%'
      AND degree ILIKE 'phd'
      AND program ILIKE '%computer science%'
      AND (program ILIKE '%georgetown%'
            OR program ILIKE '%massachusetts institute of technology%'
            OR program ~* '\\ymit\\y'
            OR program ILIKE '%stanford%'
            OR program ILIKE '%carnegie mellon%');
```

What the query does and why it answers the question

Applies all five restrictions at once (term, acceptance, PhD, Computer Science, one of the four universities) using the original downloaded program text. MIT is matched both by its full name and by the standalone word 'MIT'.

Question 9: Same question using the LLM-generated fields

Question

Repeating Question 8 but identifying the program and university with llm_generated_program and llm_generated_university, how do the counts compare?

Result

Original-field count: 30

LLM-field count: 30

Difference: 0

SQL query

```
SELECT
  COUNT(*) FILTER (WHERE program ILIKE '%computer science%'
    AND (program ILIKE '%georgetown%'
      OR program ILIKE '%massachusetts institute of technology%'
      OR program ~* '\ymit\y'
      OR program ILIKE '%stanford%'
      OR program ILIKE '%carnegie mellon%')) AS original_field_count,
  COUNT(*) FILTER (WHERE llm_generated_program ILIKE '%computer science%'
    AND (llm_generated_university ILIKE '%georgetown%'
      OR llm_generated_university ILIKE '%massachusetts institute of technology%'
      OR llm_generated_university ~* '\ymit\y'
      OR llm_generated_university ILIKE '%stanford%'
      OR llm_generated_university ILIKE '%carnegie mellon%')) AS llm_field_count
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
  AND status ILIKE 'accept%'
  AND degree ILIKE 'phd';
```

What the query does and why it answers the question

Term, degree and status still come from the original fields; only the program and university tests switch to the LLM-generated columns. One query returns both counts (the original-field count equals Question 8) and the difference is LLM minus original.

Original question 1: Fall 2026 acceptance rate by nationality group

Question

For Fall 2026, how does the acceptance rate differ between American, International and Other applicants?

Result

American: 36.73% accepted (n = 17,726)

International: 34.25% accepted (n = 14,791)

Unknown: 53.32% accepted (n = 572)

Other: 54.24% accepted (n = 118)

SQL query

```
SELECT COALESCE(NULLIF(TRIM(us_or_international), ''), 'Unknown') AS nationality_group,
  COUNT(*) AS entries,
  100.0 * COUNT(*) FILTER (WHERE status ILIKE 'accept%') / COUNT(*) AS acceptance_percent
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
GROUP BY nationality_group
ORDER BY entries DESC, nationality_group;
```

What the query does and why it answers the question

Groups Fall 2026 entries by nationality classification (blank shown as Unknown) and reports how many entries each group has and what percentage of them were acceptances.

Original question 2: Accepted versus rejected applicants' GPA and GRE

Question

Among Fall 2026 applicants who were accepted or rejected, how do average GPA and GRE Quantitative scores compare, and how many applicants report them?

Result

Accepted (n = 11,945): average GPA 3.76, average GRE Quantitative 165.61

Rejected (n = 15,084): average GPA 3.78, average GRE Quantitative 165.59

SQL query

```
SELECT CASE WHEN status ILIKE 'accept%' THEN 'Accepted' ELSE 'Rejected' END AS outcome,  
       COUNT(*) AS entries,  
       AVG(gpa) AS average_gpa,  
       AVG(gre) AS average_gre_quantitative  
FROM applicants  
WHERE LOWER(TRIM(term)) = 'fall 2026'  
      AND (status ILIKE 'accept%' OR status ILIKE 'reject%')  
GROUP BY outcome  
ORDER BY outcome;
```

What the query does and why it answers the question

Groups Fall 2026 accepted and rejected entries and averages GPA and GRE Quantitative within each group, ignoring missing values. It shows how self-reported scores differ by outcome.